

CAT1 Remote Water Meter Communication Protocol

V1.0

I. Communication Data Format

The CAT 1 communication data packet format is

as follows: Sequence	Number Name	1 byte	illustrate
1	Start Frame		0x68
2	Table Type Table Address Device	1	Water meter 0x10
3	Type	7	14-digit serial number, BCD code, low byte first.
4	Instruction Number	1	0x03
5	Control Code	2	The command number issued by the server, which defaults to 0x0000 when actively reported.
6	Spare Data	1	See control code description
7	Length	2	0x0000
8	Data Field Check	2	The length of the data field, with the high byte first.
9	Code End	n	0x255
10	Frame	1	Calculation of all data from the start frame 0x68 to the checksum
11		1	0x16

1. Start Frame

This indicates the start of a communication data packet, and is fixed as data 0x68.

2. Table type

表 3 仪表类型及其代码

仪表类型	代码(T)	仪 表
10 H~19 H:水表	10 H	冷水水表
	11 H	生活热水水表
	12 H	直饮水水表
	13 H	中水水表
20 H~29 H:热量表	20 H	热量表(计热量)
	21 H	热量表(计冷量)
30 H~39 H:燃气表	30 H	燃气表
40 H~49 H:其他仪表	40 H	如:电度表

According to the CJ-T188-2004 specification, set the meter type and its code, and set the water meter as 10H.

3. Equipment type

Fixed at 0x03.

4. Instruction Number

The number is generated by the server and sent along with the instruction to distinguish different instructions;

The numbers within each instruction must not be repeated;

The response to the instruction returned by the table carries the same number, so that the response corresponds one-to-one with the instruction;

5. Table address

The table address is represented by 7 bytes (A0~A6), with the high byte first, in 14-bit BCD code format. Its structure is shown in the table below.

Table address													
A6		A5		A4		A3		A2		A1		A0	
8	C1	C2	C3	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
Fixed value		manufacturer code			Customer customization								

6. Control Code

Control codes indicate the operation type and communication direction of communication data packets. See the table below for a description of control codes.

control code	illustrate
97H	The meter actively reports
17H	Server response to the table report
01H	The server issues a read command to the table.
81H	The response from the client to the server's read command.
04H	The server issues a write command to the table.
84H	The client's response to the server's write command.

After the server issues a read/write command to the table, the table must send the corresponding response data to the server.

7. Data length

Within a communication data packet, the data field consists of two bytes, with the most significant byte first.

8. Data Domain

The data field contains parameters and data written to or returned by the table. Its contents are as follows.

Data domain									
Parameter 1					parameter n				
1	2		n		1	2			n
Data content					Data content				

9. Verification code

The hexadecimal representation of all bytes from the start frame (68H) to the checksum (excluding the checksum) is incremented byte by byte, with only the lower 8 bits retained.

After receiving the data packet, the server and the table need to verify the data. If the verification fails, the data is considered invalid.

10. End Frame

This indicates the end of a data packet and is fixed at 0x16.

Note: Numbers in this agreement are presented in two formats: numbers followed by "H " or numbers preceded by "0x " to indicate hexadecimal numbers, without "H ".

And "0x " is a decimal number.

II. Communication Commands

1. Server table read parameters

After the meter is online, the server can issue commands to read the meter's parameters. The specific protocol format is as follows:

1.1 Table Reading Parameters

The server issues a command.

Serial Number		length	Byte position	Code
1	Description:	1	0	68H
2	Start Frame,	1	1	T
3	Table Type,	7	2—8	A6—A0
4	Table Address,	1	9	03H
5	Device Type,	1	10	01H
6	Control Code,	2	11—12	xxxxH
7		2	13—14	00H
8	Instruction Number,	1	15	m
9	Spare Data	2	16—17	A901H
10	Length,	8	18—25	
11	Data Identifier,	1	26	CS
12	Spare Check Code, End Frame	1	27	16H

Response

Serial Number		length	Byte position	Code
1	Description	1	0	68H
2	of start frame	1	1	T
3	table type	7	2—8	A6—A0
4	table address	1	9	03H
5	device type	1	10	81H
6	control code	2	11—12	xxxxH
7		2	13—14	00H
8	instruction number	1	15	
9	spare data length data identifier	2	16—17	A901H
10	IMEI	8	18—25	
11	IMSI	8	26—33	
12	ICCID	10	34—43	
13	Hardware version number	2	44—45	

14	Software Version	2	46—47	
15	Number Table Type	2	48—49	
16	Code Vendor	2	50—51	
17	Code Daily Limit Server	1	52	
18	Address Server	6	53—58	
19	Address Cumulative	6	59—64	
20	Reporting Count	2	65—66	
21	Reporting Mode	6	67—72	
22	Valve Control Function	2	73—74	
23	Shield Metering	1	75	
24	Mode Payment	1	76	
25	Mode In Place	1	77	
27	Mode Current	1	78	
28	Tier	1	79	Step-by-step effective
29	Settlement	1	80	
30	Monthly	2	81—82	
31	Settlement	2	83—84	
32	Day Tier	2	85—86	
33	Quantity 1 Tier	2	87—88	
34	Quantity 2 Tier	2	89—90	
35	Quantity 3 Tier	2	91—92	
36	Price 1 Tier Price	2	93—94	
37	2 Tier Price 3 Tier Price	2	95—96	
40	4 High Flow Threshold 38 High Flow Threshold	1	97	
	Time Low Flow	2	98—99	
40	Threshold 39 Low Flow Threshold Time	1	100	
41	Minimum Water Quantity/	1	101	Unit: cubic meters/yuan
42	Amount Extra Water	1	102	
43	Quantity/Amount	1	103	
44	Alarm Quantity/	1	104	
45	Amount	8	105—112	
46	Overdraft	1	113	CS
47	Quantity/Amount Backup Verification Code End Frame		114	16H

2. Server write table parameters

After the meter is online, the server can issue commands to modify the meter's parameters. The meter responds to the command and sends back response data. The specific protocol format is as follows:

Server -> Table:

Serial Number		length	byte position	Code
1	Description	1	0	68H
2	of start frame	1	1	T
3	table type	7	2—8	A6—A0
4	table address	1	9	03H
5	device type	1	10	04H
6	control code	2	11—12	xxxxH
7		2	13—14	0000H
8	instruction number	1	15	m
9	spare data length data identifier	2	(15+1)—(15+2)	xxxxH
10	Parameter 1 Data		(15+3)—	
11	—	—	—	
12	Parameter n: Data		—y15+mÿ	
13	check code	1	15+m+1	CS
14	end frame	1	15+m+2	16H

Table -> Server: Serial

Number		length	byte position	Code
1	Description	1	0	68H
2	of start frame	1	1	T
3	table type	7	2—8	A6—A0
4	table address	1	9	03H
5	device type	1	10	84H
6	control code	2	11—12 13—	xxxxH
7		2	14 15	0000H
8	instruction number	1		11H
9	spare data length data identifier	2	16—17	xxxxH
10	Success sign	1	18	0 indicates success; non-zero indicates error. code
11	Real-time clock	6	19—24	
12	backup	8	25—32	
13	check code	1	33	CS
14	end frame	1	34	16H

2.1 Power-off procedure

Serial Number	Explain	length	Byte position	Code
1	the type of the	1	0	68H
2	start frame table	1	1	T
3	Table address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	control code	1	10	04H
6	Instruction Number	2	11—12	xxxxH
7	spare	2	13—14	0000H
8	Data length	1	15	11H
9	Data Identifier	2	16—17	AA00H
10	Clock calibration enable	1	18	0x5A Clock Calibration
11	real-time clock	6	19—24	Year Month Day Hour Minute Second BCD
12	backup	8	25—32	
13	Verification code	1	33	CS
14	End frame	1	34	16H

2.2 Initialization Operation

none

2.3 Reset Operation

none

2.4 Accumulation Operation

none

2.5 Sales operations are not
included.

2.6 Valve Operation

Serial Number	Explain	length	Byte position	Code
1	the type of the	1	0	68H
2	start frame table	1	1	T
3	Table address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	control code	1	10	04H
6	Instruction Number	2	11—12	xxxxH
7	spare	2	13—14	0000H
8	Data length	1	15	0CH
9	Data Identifier	2	16—17	AA05H
10	Valve opening/	1	18	0x55 On, 0x99 Off
11	closing operation	1	19	0x5A is mandatory, others are not.
12	type: spare	8	20—27	
13	Verification code	1	28	CS
14	End frame	1	29	16H

2.7 Setting Reporting/Valve Control Parameters

Serial Number	illustrate	length	Byte position	Code
1	Start Frame	1	0	68H
2	Table Type	1	1	T
3	Table Address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	Control code	1	10	04H
6	instruction number	2	11—12	xxxxH
7	spare	2	13—14	0000H
8	Data length, data	1	15	28
9	identifier, daily	2	16—17	AA06H
10	limit on the number of times	1	18	3~100, default 0
11	Reporting mode	6	19—24	
12	valve control function shielding	2	25—26	
13	Historical data enabled	1	27	0xAC enabled 0xCA Close
14	Daily limit on	6	28-33	
15	emergency reporting times modified	1	34	0xAD allows modification
16	Valve control function modification	1	35	0xAE allows modification
17	Backup	8	36—43	
18	check code	1	44	CS
19	end frame	1	45	16H

2.8 Setting the Meter Type

Serial Number		length	Byte position	Code
1	Description	1	0	68H
2	of start frame	1	1	T
3	table type	7	2—8	A6—A0
4	table address	1	9	03H
5	device type	1	10	04H
6	control code	2	11—12	xxxxH
7		2	13—14	0000H
8	instruction number spare data length	1	15	0DH
9	Data Identifier	2	16—17	AA07H
10	Metering mode	1	18	
11	Paid model	1	19	
12	In place mode	1	20	
13	Address modification enabled	1	21	0xC1 Allows modification of table number
14	New address	7	22—28	
15	Manufacturer modification enable	1	29	0xC3 allows modification of the manufacturer.
16	New	2	30—31	
17		8	32—39	
18	manufacturer's	1	40	CS
19	backup check code end frame	1	41	16H

2.9 Setting Meter Parameters

Serial Number	Description	length	Byte position	Code
1	of the start frame	1	0	68H
2	Table type	1	1	T
3	Table address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	control code	1	10	04H
6	Instruction Number	2	11—12	xxxxH
7	spare	2	13—14	0000H
8	Data length	1	15	43
9	Data Identifier	2	16—17	AA08H
10	Minimum water volume/amount	1	18	Unit: cubic meters/yuan
11	Additional water/amount	1	19	
12	Alarm quantity/amount	1	20	
13	Overdraft Amount/Amount	1	21	

14	Settlement Month	1	22	Step-by-step effective
15	Settlement date	1	23	
16	Step quantity 1	2	24—25	
17	Step quantity 2	2	26—27	
18	Step quantity 3	2	28—29	
19	Tiered pricing 1	2	30—31	
20	Tiered pricing 2	2	32—33	
21	Tiered pricing 3	2	34—35	
22	Tiered pricing 4	2	36—37	
23	High flow threshold	2	38—39	
24	High flow threshold time	1	40	
25	Low flow threshold	2	41—42	
26	Low flow threshold time	1	43	
27	High flow allow settings	1	44	0xA9 is allowed to be set; others are invalid.
28	Small flow allow settings	1	45	0xAA is allowed; others are invalid.
29	Step parameters can be set	1	46	0xAB is allowed; others are invalid.
30	Minimum Guarantee Settings	1	47	0xAC is allowed; others are invalid.
31	Additional allowed settings	1	48	0xAD is allowed to be set; others are invalid.
32	Alarm Allow Settings	1	49	0xAE is allowed; others are invalid.
33	Overdraft allowance settings	1	50	0xAF is allowed; others are invalid.
34	spare	8	41—58	
35	Verification code	1	59	CS
36	End frame	1	60	16H

2.10 Setting the Base Number of Postpaid Tables

Serial Number	illustrate	length	Byte position	Code
1	Start Frame	1	0	68H
2	Table Type	1	1	T
3	Table Address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	Control code	1	10	04H
6	instruction number	2	11—12	xxxxH
7	spare	2	13—14	0000H
8	Data length, data	1	15	0EH
9	identifier, table	2	16—17	AA09H
10	base number,	4	18—21	
11	spare	8	22—29	
12	check code,	1	30	CS
13	end frame.	1	31	16H

3. Table actively uploads data

Client-side -> Server:

Serial Number		length	Byte position	Code
1	Description	1	0	68H
2	of start frame	1	1	T
3	table type	7	2—8	A6—A0
4	table address	1	9	03H
5	device type	1	10	97H
6	control code	2	11—12	0000H
7	instruction number	2	13—14	xxxxH
8	reporting type	1	15	m
9	data length data packet type	1	15+1	xxH
10	Parameter 1 Data		(15+2)—	
11	—	—	—	
12	Parameter n: Data		—y15+mȳ	
13	checksum	1	15+m+1	CS
14	end frame	1	15+m+2	16H

Server -> Table: Serial Number 1

		length	Byte position	Code
	Description	1	0	68H
2	of Start	1	1	T
3	Frame, Type	7	2—8	A6—A0
4	Table, Address,	1	9	03H
5	Device Type,	1	10	17H
6	Control Code,	2	11—12	0000H
7	Instruction	2	13—14	xxxxH
8	Number, Reporting	1	15	m
9	Type, Data Length,	1	16	xxH
10	Data Packet	1	17	0xAF is valid
11	Type, Power	1	18	CS
12	Off Flag, Checksum, End Frame	1	19	16H

The instruction number in the active reporting and server response packets is set to 0x0000;

After receiving the information proactively reported by the client, the server needs to issue a response command;

If no further instructions are received, the power-off flag is set to 0xAF, and the device will power off immediately upon receiving the command.

If there are subsequent instructions, the shutdown flag will be set to 0x00, and the terminal will extend the waiting time to wait for instructions from the server.

Data packet type:

- 01—Prepaid General Form Reporting Information
- 02—Prepaid tiered table reporting information
- 03—Postpaid Standard Form Reporting Information
- 04—Information reported by general HVAC valve meters
- 05—Reporting Information on Additional Charges/Minimum Consumption
- 06 - Daily Historical Data Reporting Information

Reporting type:

BIT	D15	D14	D13	D12	D11	D10	D9	D8
definition	- Data retransmission and intensive sampling traffic exceeded limits				Daily History data	Overdraft alarm and remaining balance alarm		
BIT	D7	D6	D5	D4	D3	D2	D1	D0
Define undervoltage alarm backup upload			Additional/Guarantee Bottom deduction	Magnetic interference valve action		card swipe reporting,	trigger reporting,	timed reporting

3.1 Uploading data packets to a regular table 01

This data package contains the daily reported data for a standard prepaid meter. It is also reported when the valve is activated, activated by a button touch, or activated by a card swipe.

According to the content.

Serial Number		length	Byte position	Code
1	Explanation	1	0	68H
2	of Start	1	1	T
3	Frame Table	7	2—8	A6—A0
4	Type Table	1	9	03H
5	Address	1	10	97H
6	Device Type	2	11—12	0000H
7	Control Code	2	13—14	xxxxH
8	Instruction	1	15	m
9	Number Reporting	1	16	01H
10	Type Data Length	2	17—18	
11	Data Packet Type	2	19—20	xx
12	Manufacturer Code	2	21—22	
13	Table Type Code Hardware Version Software Version	2	23—24	
14	IMEI	8	25—32	
15	ICCID	10	33—42	
16	voltage	2	43—44	
17	signal strength,	1	45	
18	signal quality,	1	46	
19	signal-to-noise ratio	1	47	

20	Cumulative number of	2	48—49	
21	reports, daily number of	2	50—51	
22	reports, reporting	6	52—57	
23	mode, status	2	58—59	
24	word, number of	1	60	
25	purchases,	4	61—64	
26	cumulative	4	65—68	
27	usage, remaining	4	69—72	
28	amount, total	4	73—76	
29	purchase	4	77—80	
30	amount, daily	6	81—86	
31	usage,	1	87	CS
32	monthly usage, real-time clock, checksum, end frame.	1	88	16H

3.2 Upload data package 02 for prepaid tiered table

This data package contains the daily reported data for the tiered prepaid meter. It is also reported when the valve is activated, activated by a button touch, or activated by a card swipe.

content.

Serial Number	illustrate	length	byte position	Code
1	Start Frame	1	0	68H
2	Table Type	1	1	T
3	Table Address	7	2—8	A6—A0
4	Equipment type	1	9	03H
5	control code	1	10	97H
6	instruction number	2	11—12	0000H
7	Reporting type	2	13—14	xxxxH
8	Data length	1	15	78
9	Data packet type	1	16	02H
10	Manufacturer code	2	17—18	
11	Table type code	2	19—20	xx
12	hardware version	2	21—22	
13	Software version	2	23—24	
14	IMEI	8	25—32	
15	ICCID	10	33—42	
16	Voltage	2	43—44	
17	signal strength,	1	45	
18	signal quality,	1	46	
19	signal-to-noise	1	47	
20	ratio, cumulative reporting times	2	48—49	

21	Daily Reporting Count	2	50—51	
22	Reporting Mode	6	52—57	
23	Status	2	58—59	
24	Purchase Count	1	60	
25	Cumulative	4	61—64	
26	Amount	4	65—68	
27	Remaining Amount	4	69—72	
28	Total	4	73—76	
29	Purchase	4	77—80	
30	Amount Daily	4	81—84	
31	Usage Monthly	1	85	
32	Usage Cumulative Usage	2	86—87	
33	Current Tier	6	88—93	
34	Current Tier Price Real-time Clock Verification Code		94	CS
35	End frame	1	95	16H

3.3 Postpaid Table Upload Data Package 03

This data package contains the daily reported data for a standard postpaid meter. It is also reported when the valve is activated, activated by a button touch, or activated by a card swipe.

Contents: Serial Number		length	byte position	Code
1	Description	1	0	68H
2	of start	1	1	T
3	frame table	7	2—8	A6—A0
4	type table	1	9	03H
5	address	1	10	97H
6	device type	2	11—12	0000H
7	control code	2	13—14	xxxxH
8	instruction	1	15	m
9	number reporting type data length data packet type	1	16	03H
10	Manufacturer code	2	17—18	
11	Table type code	2	19—20	
12	hardware version	2	21—22	
13	Software version	2	23—24	
14	IMEI	8	25—32	
15	ICCID	10	33—42	
16	voltage	2	43—44	
17	signal strength	1	45	
18	and signal quality	1	46	

19	Signal-to-	1	47	
20	noise ratio cumulative	2	48—49	
21	reporting count, daily	2	50—51	
22	reporting count,	6	52—57	
23	reporting	2	58—59	
24	mode, status word,	4	60—63	
25	cumulative	4	64—67	
26	usage, daily	4	68—71	
27	usage, monthly	6	72—77	
28	usage, real-	1	78	CS
29	time clock checksum, end frame.	1	79	16H

3.5 Upload data packets to a regular table 05

This data package contains a prepaid billing table (regular prepaid and tiered prepaid), which sets minimum spending, guaranteed minimum spending, and additional charges. When a charge occurs, the data is proactively reported.

According to the content. Serial number.		length	byte position	Code
1	Description	1	0	68H
2	of start	1	1	T
3	frame table	7	2—8	A6—A0
4	type table	1	9	03H
5	address	1	10	97H
6	device type	2	11—12	0000H
7	control code	2	13—14	xxxxH
8	instruction	1	15	m
9	number reporting type data length data packet type	1	16	05:00
10	Manufacturer code	2	17—18	
11	Table type code	2	19—20	xx
12	hardware version	2	21—22	
13	Software version	2	23—24	
14	Additional charges/amount;	4	25—28	points/L
15	minimum charge/amount; voltage; signal	4	29—32	points/L
16	strength;	2	33—34	
17	signal quality;	1	35	
18	signal-to-noise	1	36	
19	ratio;	1	37	
20	cumulative number of	2	38—39	
21	reports; daily reports; status	2	40—41	
22	word	2	42—43	
23	purchases;	1	44	
24	cumulative usage/amount.	4	45—48	points/L

25	Remaining quantity/amount	4	49—52	points/L
26	Total purchase quantity/amount	4	53—56	points/L
27	Verification	1	57	CS
28	code End frame	1	58	16H

3.6 Table upload date 48:00 cumulative usage data package 06

This data packet contains prepaid tables (regular prepaid and tiered prepaid) and postpaid tables, and actively reported data, including date, 48-point data, etc.

Serial Number		length	byte position	Code
1	Description	1	0	68H
2	of start	1	1	T
3	frame table	7	2—8	A6—A0
4	type table	1	9	03H
5	address	1	10	97H
6	device type	2	11—12	0000H
7	control code	2	13—14	xxxxH
8	instruction	1	15	216
9	number reporting type data length data packet type	1	16	06H
10	Manufacturer code	2	17—18	
11	Table type code	2	19—20	xx
12	hardware version	2	21—22	
13	Software version	2	23—24	
14	Voltage	2	25—26	
15	signal strength,	1	27	
16	signal quality,	1	28	
17	signal-to-	1	29	
18	noise ratio, cumulative	2	30—31	
19	number of reports, daily	2	32—33	
20	report count,	2	34—35	
21	status (date),	4	36—39	Example: 20231011
22	00:29, cumulative water volume.	192	40—231	Unit: L
23	00:59 Cumulative water volume			
24	01:29 Cumulative water volume			
25	...			
26	Cumulative water volume at 23:29			
27	End of	1	232	CS
28	checksum frame	1	233	16H

III. Parameter Description

The parameters used by the system are shown in the table below.

	Serial Number	Data Identifier	Parameter Name	Format	Length	Permissions	0xAB00 Table Number	Hex	illustrate
equipment information	1						7	R	is the table name, a 14-digit number.
	2	0xAB01	IMEI	BCD			8	R	module IMEI, 15-digit BCD code, padded with zeros at the lowest level.
	3	0xAB02	IMSI	BCD			8	R	Card Number IMSI, 15-digit BCD code, padded with zeros at the lowest level.
	4	0xAB03	ICCID	BCD			10	R	Card number ICCID, 20-digit BCD code
	5	0xAB04	Hardware version number	HEX			2	R	hardware circuit version
	6	0xAB05	Software version number	HEX			2	R	is the program version that was burned.
	7	0xAB06	Table type code	HEX	0xAB07		2	R	varies depending on the set criteria, payment method, etc.
	8	Vendor code	HEX 0xAC00	Status word			2	R	such as 0xC1F1
Mounting state	1	HEX Spare	0xAC01				2	R	(See instructions)
	2			HEX			1	R	——
	3	0xAC02	Battery Voltage	HEX	Water		2	R	is hexadecimal, unit: mV
	4	0xAC03	Pressure	HEX			1	R	(Reserve)
	5	0xAC04	Temperature	HEX			1	R	(unit: °C)
	6	0xAC05	Current Tier	HEX 0xAC06	Current		1	R	ranges from 1 to 3
	7	Price	HEX 0xAC07	Signal Quality	HEX		2	R	(unit: minutes/m³)
	8	0xAC08	Signal Strength	HEX 0xAC09			1	R	RSRQ, with a sign, such as -68dBm
	9	Signal-to-Noise Ratio	HEX 0xAC0A	Signal			1	R	RSRP
	10	Power	HEX 0xAC0B	Cumulative Reporting			1	R	SNR, with a sign, such as -68dBm
	11	Count	HEX 0xAC0C	Daily Reporting Count			1	R	RSRP, with a sign, such as -1, +15
	12	HEX 0xAC0D	Daily Limit Count	HEX			2	R	
	13	0xAC0E	Server Address 1	HEX 0xAC0F			1	R	
	14	Server Address 2	HEX 0xAC10	Reporting			1	R/W	
	15	Mode	HEX 0xAC11	Emergency Reporting			6	R/W	server: 4-byte address + 2-byte port number
	16	HEX 0xAC12	Real-Time Clock	BCD			6	R/W	server: 4-byte address + 2-byte port number
	17	0xAD00	Valve Control Function	Shielding			6		See instructions for R/W and upload time information.
	18	HEX					6		See instructions for R/W, and the emergency reporting interval/number of times.
	19						6	R/W	Year Month Day Hour Minute Second
Mounting parameter	1						2	R/W	see instructions
	2	0xAD01	Metering Mode	HEX			1	R/W	0x50-1L 0x60-10L 0x70-100L 0x80-1000L
	3	0xAD02	Paid Mode	HEX			1	R/W	0x48 - Postpaid 0x59 - Prepaid 0x4A - Pre-step 0x4E - HVAC Valve

	4	0xAD03 Bit-in mode HEX		1	R/W	0x44 - Stalled 0x4B - Switch
	5	0xAD04 Tiered Settlement Monthly HEX		1	R/W	The value ranges from 0 to 12, with 0 representing the monthly step size; [other values are not specified]. 0-year ladder
	6	0xAD05 Tiered Settlement Date (HEX) 0xAD06 Tiered		1	R/W	ranges from 1 to 28.
	7	Quantity 1 0xAD07 Tiered Quantity 2	HEX	2	R/W unit:	m³
	8	0xAD08 Tiered Quantity 3 0xAD09	HEX	2	R/W unit:	m³
	9	Tiered Price 1 (HEX) 0xAD0A Tiered	HEX	2	R/W unit:	m³
	10	Price 2 0xAD0B Tiered Price 3		2	R/W unit:	minutes/m³
	11		HEX	2	R/W unit:	minutes/m³
	12		HEX	2	R/W unit:	minutes/m³
	13	0xAD0C Tiered pricing 4 HEX 0xAD0D		2	R/W unit:	minutes/m³
	14	High flow threshold HEX High flow		2	R/W unit:	L/minute
	15	0xAD0F threshold time between	HEX	2	R/W (unit:	minutes)
	16	0xAD10 Low flow threshold value HEX Low flow		2	R/W unit:	L/minute
	17	0xAD11 threshold time between	HEX	2	R/W (unit:	minutes)
	18	0xAD12 Guaranteed Water Supply (HEX) 0xAD13		1	R/W unit:	cubic meters
	19	Guaranteed Amount (HEX) 0xAD14 Extra Water		1	R/W Unit:	Yuan
	20	Supply (HEX) 0xAD15 Extra Amount (HEX) 0xAD16		1	R/W unit:	cubic meters
	21	Alarm Amount (HEX) 0xAD17 Alarm Amount (HEX)		1	R/W Unit:	Yuan
	22	0xAD18 Overdraft Amount (HEX) 0xAD19 Overdraft		1	R/W unit:	cubic meters
	23	Amount (HEX) 0xAD20 Delay Days (HEX) 0xAD21		1	R/W Unit:	Yuan
	24	Historical Data Function (HEX) 0xAE00 Number of		1	R/W unit:	cubic meters
	25	Purchases (HEX) 0xAE01 Total Purchases (HEX)		1	R/W Unit:	Yuan
	26	0xAE02 Total Purchase Amount (HEX) 0xAE03		1	R/W unit:	days
	27	Cumulative Usage (HEX) 0xAE04 Cumulative Amount		1	R/W	0xAC enabled; 0xCA disabled; others invalid.
user data	1	(HEX) 0xAE05 Remaining Amount (HEX) 0xAE06		1	R	takes values from 0 to 255; values exceeding 255 become 1.
	2	Remaining Amount (HEX) 0xAE07 Daily Usage (HEX)		4	R (Unit:	L)
	3	0xAE08 Monthly Usage (HEX) 0xAE09 Monthly Usage		4	R Unit:	minutes
	4	Amount (HEX)		4	R/W Unit:	L (Postpaid meters can be configured with a base value).
	5			4	R Unit:	minutes
	6			4	R (Unit:	L)
	7			4	R Unit:	minutes
	8			4	R (Unit:	L)
	9			4	R (Unit:	L)
	10			4	R Unit:	minutes
	11	0xAE0A Historical Data 1 HEX 196			R	Use once every half hour, for a total of 2*24 sets. Unit: L
	12	0xAE0B Historical Data 2 HEX 196			R	Use once every half hour, for a total of 2*24 sets. Unit: L

	13	0xAE0B Historical Data 3 HEX 196			R	Use once every half hour, for a total of 2*24 sets. Unit: L
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1. 0xAA04 – Sales Operation

Phenotype		Byte 1 - Byte 4: Sales/Purchase Quantity/	5th byte
water meter	definition	Amount	Purchase count
	illustrate	Unit: L/min	1~255 If the value exceeds 255, it will automatically change to 1.

2. 0xAA05 — Valve Operation

Phenotype		Byte 1	2nd byte
water meter	definition	Valve operation types	Operation permissions
	illustrate	0x55: Open valve 0x99: Valve closed	0x5A: Forced Operation !0x5A: Non-mandatory operation

3. 0xAB06 — Table type code

Phenotype		D15-D8	D7	D6	D5	D4	D3	D2	D1	D0
water meter	Definition	Payment Type	Payment Type	Table Scope						
	illustrate	—	—	—	00: Prepaid 01: Pre-ladder 10: Postpaid 11: HVAC valve		—	0: Blocked turn 1: Switch	00ÿ 1L 01ÿ 10L 10ÿ100L 11:1000L	

4. 0xAB07 — Manufacturer Code

Phenotype		D15-D8	D7-D0
water meter		Manufacturer's high-order	Manufacturer low byte
	Definitions	byte example: 0xC1	Example: 0xF1

5. 0xAC0E – Server Address Description

Phenotype		B0	B1	B2	B3	B4	B5
water meter	definition	IP address				port number	
	For example	0xC0	0xA8	0x01	0x64	0x2766	
	illustrate	192	168	1	100	10086	

It contains a 4-byte IP address, a 2-byte port number, and the high byte of the data is listed first.

IP address: 192.168.1.100 Port number: 10086

6. 0xAC00 — Status word definition

Phenotype		D7	D6	D5	D4	D3	D2	D1	D0
water meter	Define the battery voltage with magnetic interference when the cover is opened						— — —	Valve status	
	illustrate	0: Normal 1: Open the lid	0: Normal 1: Interference	0: Normal 1: Undervoltage				00: Open 01: Guan 11: Abnormal	

Phenotype		D15	D14	D13	D12	D11	D10	D9	D8
water meter	Definition	Magnetic Interference Record		Card Swipe Record					
	illustrate-		0: Normal 1: Interference		0 - Normal 1-Valve opening card	0 - Normal 1-Valve opening card			

7. 0xAC10 — Reporting Mode

Phenotype		B0	B1	B2	B3	B4	B5
water meter	Upload interval in minutes 0xC0		Interval minutes high byte	Interval minutes low byte			
	Specify multiple dates: 0xC1 Date1 Date2 Date3					Date 4 hours	
	Specify multiple hours: 0xC2, 1 hour, 2 hours, 3 hours, 4 minutes.						
	Specify multiple minutes: 0xC 3 minutes 1 minute 2 minutes 3 minutes 4 minutes 5 minutes						

Default value: 0xC005A0000000, reported every 1440 minutes.

Note: The interval between the high and low bytes of minutes is in hexadecimal format;

The remaining time data is in BCD format;

8. 0xAC11 — Emergency Reporting

This feature can only be enabled for prepaid models;

When a prepaid meter is shut down due to unpaid fees, and the emergency reporting function is enabled, it can report information and receive instructions at intervals.

If the number of emergency reports exceeds the set limit, reporting will stop.

If a valve opening instruction (sales order/valve opening instruction) is received during the emergency reporting process, the reporting should be stopped.

When the valve is shut off again due to non-payment, the emergency reporting function can be triggered again.

Phenotype	B0	B1	B2	B3	B4	B5
water meter	0x5AA5 Enabled 0xA55A Disabled Other invalid		High-resolution text at intervals of minutes Festival	Interval in minutes byte	High maximum number of reports byte	Maximum number of reports Count low byte

Default value: 0x00, not enabled.

Note: The interval between the high and low bytes of minutes is in hexadecimal format;

The interval minute setting range is 20~720 minutes;

9. 0xAD00 — Valve-controlled function disabled

Valve Control Function Shielding Table

Phenotype		D15-D8 D7		D6	D5	D4	D3	D2	D1	D0
water meter	definition	—	Opening the cover causes magnetic interference and low-voltage power loss in the battery.				—	—	—	—
	illustrate-		0: Valve closed 1: Invalid	0: Valve closed 1: Invalid	0: Valve closed 1: Invalid	0: Valve closed 1: Invalid	— — —			

Default value: 0x0000, valve control is enabled.